Certifying Voltage Headroom on AI-Loaded Distribution Feeders: Closed-Form and Neural Band Guarantees Toward a Decarbonized Grid

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Abstract

AI-training workloads concentrate into campuses whose active power steps by tens of percent every few seconds as the training loop alternates between compute and communication phases; on a feeder sized for slowly varying load, that square wave swings bus voltages across the $\pm 5\%$ interconnection band. We verify a switching-reference droop on an IEEE 33-bus feeder: within a workload mode the closed loop is the linear system $e_{t+1} = (I - XK)e_t + d_t$, whose safety is a spectral radius and a convergent matrix sum. We certify that loop ($\rho(I - XK) = 0.99941$) and a neural replacement, proving with CROWN and alpha-CROWN that every state in an operating box and every data-center load in the mode range stays in-band, with certified one-step contraction $\gamma \approx 0.61$. Priced honestly—0.042 certified (alpha-CROWN) against 0.149 empirical, an IBP that certifies nothing—it turns a sampled confidence score into a sound radius an interconnection study can sign.

1 Introduction

A training run and the feeder it sits on pull against each other. Compute and communication phases differ in instantaneous power by tens of percent; across thousands of accelerators that step is synchronous and repeatable on a seconds timescale, so the data center presents to its feeder the active-power load

$$p_{j,t} = \bar{p}_j(m_t) + w_t, \qquad m_t \in \{0, 1\}, \qquad |w_t| \le \bar{w}, \tag{1}$$

where m_t is the workload mode, $\bar{p}_j(\cdot)$ the mode-dependent mean load at bus j, and w_t bounded intra-mode fluctuation. A feeder is sized to the running average, not the step, so bus voltages swing at each transition and cross the $\pm 5\%$ band that interconnection agreements are written against. The genuinely fast local lever is reactive power: on a radial line the voltage is an affine function of the power injected along the root-to-bus path, so a reactive source at the data center can push it back. Volt/VAR control is thus a per-bus droop that must hold the band while the load moves in seconds, and an interconnection study must be able to *guarantee* the band, which a sampled confidence score cannot do.

A fixed-reference droop fails structurally: during compute it winds up reactive power to hold the sagging voltage near nominal, then over-corrects past $+5\%$ at every transition (Figure 4a, App. A). The switching-reference controller of Yan et al. [1] fixes this by switching the voltage reference with the workload mode (Section 2), and a ReLU-network replacement for it is not a linear map but is globally piecewise affine—exactly the structure that bound-propagation verifiers [2, 3] can certify over a whole region at once. This is the same governing lesson as the certified-neural-control line it builds on [4, 5], namely that near-perfect sampled satisfaction still admits counterexamples; no claim here rests on sampling. Yan et al. [1] propose the control law this paper certifies but claim no proof

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over an operating envelope; that proof—for their controller and for a learned replacement—is our contribution.

Contributions. An exact, closed-form certificate for the published switching-reference controller as a spectral radius and matrix sum on the real feeder ($\rho(I - XK) = 0.99941$, Section 2, Table 2); a neural controller in the same loop certified by CROWN and alpha-CROWN to keep every bus in-band for all states in an operating box and all loads in the mode range, with certified contraction $\gamma \approx 0.61$ (Section 3.2); and an honest account of the price of soundness, including a vacuous 252 p.u. scalar bound and the 0.042-versus-0.149 boundary (§4).

2 Problem formulation and control law

We model the feeder by LinDistFlow [6]; near 1 p.u. the voltage profile is the affine map

$$v = Rp + Xq + \not\Vdash, \tag{2}$$

where $v, p, q \in \mathbb{R}^n$ are the non-slack bus voltages and the active and reactive power injections in per unit, and R, X are symmetric positive-definite path-overlap sensitivity matrices (App. A); we use the real Baran-Wu IEEE 33-bus feeder (App. A), and m_t steps every $60/20$ steps (App. D). All controllers are decentralized per-bus reactive droops with a rate limit and capacity cap,

$$q_{t+1} = q_t - K\big(v_t - v^{\mathrm{ref}}\big), \tag{3}$$

with $K = kI$, $0 < k < 2/\lambda_{\max}(X)$. The fixed droop holds $v^{\mathrm{ref}} = 1$ and re-fights every transition. The switching reference sets v^{ref} per mode so that the mode term in the closed loop is cancelled and the two mode envelopes are centered on 1 p.u. (Props. 2–3 of Yan et al. [1]); we run it both as a model-based oracle and as an adaptive variant that infers amplitude, bias, and mode sign from the local voltage window only. Within a mode the reference is constant, so on the tracking error $e_t = v_t - v^{\mathrm{ref}}$ the loop is linear,

$$e_{t+1} = Ae_t + d_t, \qquad A = I - XK, \qquad d_t = Rp_t - v^{\mathrm{ref}}, \tag{4}$$

and the switching reference leaves $d_t = Rw_t$ with $|w_t| \le 2\bar{w}$. The safety of this loop is a spectral condition, computable in closed form with no search and no sampling:

$$0 < K < 2X^{-1} \iff \rho(I - XK) < 1, \tag{5}$$

under which the steady error is bounded by the convergent Neumann sum $S = \sum_{n \ge 0} |A^n|$: $\sup_t |e_t| \le S\bar{d}$ elementwise (App. B), where $\bar{d}$ is the per-bus disturbance bound.

3 Verification and results

3.1 The structural fix is exactly certifiable

The affine system (4) is verifiable in closed form: check (5), then form the envelope $S\bar{d}$. On the switching controller of Section 2 with $k = 0.351$, Table 2 reports the exact certificate. The gain condition holds and $\rho(I - XK) = 0.99941$; every closed-loop trajectory lies inside the per-bus envelope, which uses at most 1.26% of its own allowance (Figure 1). This is a guarantee over the continuous state space, not a sampled satisfaction fraction. The spectral value near 1 is a property of this feeder, not a red flag: its weak slow lateral mode is set by a diagonal decentralized gain, so it sits close to the unit circle, yet the certificate still bounds every trajectory. Tightening that mode is exactly what bound propagation does in Section 3.2. Two costs are real and reported: the scalar norm bound $\frac{\sqrt{n}\,C\bar{d}}{1-\rho} = 252$ p.u. is vacuous, and the envelope is loose on the slow lateral mode (Figure 1)—precisely why the neural side needs bound propagation. Table 1 and Figure 4 ground the setting: the fixed droop overshoots the band at every compute-to-communication transition, while the switching reference holds it at $8.6\times$ less reactive action in the single-DC case. In the harder two-data-center case the local-only adaptive controller grazes the band at 0.0538 on 0.006% of bus-steps; we report this rather than rounding it away.

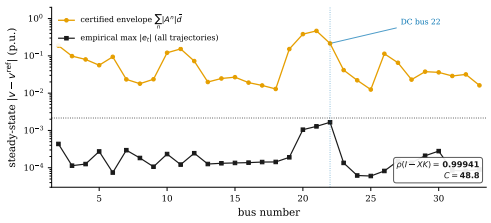


Figure 1: **The exact certificate contains every trajectory.** Certified per-bus envelope $S\bar{d}$ (orange) versus the empirical maximum over every simulated closed-loop trajectory (black markers). The envelope dominates at every bus and uses at most 1.26% of its allowance; it is loose on the slow lateral mode (far laterals), which is precisely why the neural side needs bound propagation.

Table 1: Volt/VAR control on the IEEE 33-bus feeder, mean over five seeds: peak deviation from nominal, fraction of bus-steps outside $\pm 5\%$, and cumulative reactive action $\sum |\Delta q|$ (p.u.).

Scenario	Controller	$\max \lvert v-1 \rvert$	viol.%	$\sum \lvert \Delta q \rvert$
single DC (22)	fixed droop	0.0587	0.168	13.16
single DC (22)	oracle	0.0429	0.000	0.00
single DC (22)	adaptive (ours)	**0.0439**	**0.000**	**1.53**
two DC (22,25)	fixed droop	0.0611	0.272	34.64
two DC (22,25)	oracle	0.0441	0.000	0.00
two DC (22,25)	adaptive (ours)	0.0538	0.006	8.21

3.2 A neural controller certifiable for all states

A learned reactive policy is not a linear map, but a ReLU network is globally piecewise affine, so linear-relaxation bound propagation can certify it. We build the composite one-step map

$$\mathrm{OneStep}(e, p) = Rp + X\nu(e) + Xq^{\mathrm{base}}, \tag{6}$$

the frozen physics R, X around the MLP ν. CROWN [2] and alpha-CROWN [7, 8] (via auto-LiRPA [3]) propagate sound linear bounds through ν and the linear maps, producing for every e in the box $\{e : \|e\|_\infty \le r\}$ and every p in the load range a certified box $[lb, ub]$ on the next-step error $e^+ = e_{t+1}$; the band is certified iff $[lb, ub] \subseteq [-0.05, 0.05]$. This is a guarantee over a continuum of inputs. An MLP behavior-cloned from the switching controller plus a worst-case band-excursion penalty is certified by CROWN and alpha-CROWN to hold the band for all states within $r = 0.04$ and all loads in the mode range (Figure 2, Table 3), with certified one-step contraction $\gamma = 0.614$ (compute) /0.592 (communication) at box half-width 0.02. IBP certifies nothing at any radius: its bound reaches ± 1.3 because it ignores the correlations the backward pass captures—a concrete demonstration that the method, not the architecture, makes this certifiable. Because the map also contracts the box at rate $\gamma \le 0.61$ per step, the one-step argument compounds toward forward invariance over a horizon.

4 Cost and limitations

We state the trade-offs without softening them. Soundness is priced in radius and in tightness. The certified guarantee covers a box of radius at most 0.042 (alpha-CROWN; 0.0409 for CROWN), a fraction of the true 0.149 safe set; the certified bound is about $3\times$ the empirical worst case at the headline radius; the scalar bound is 252 p.u. and unusable. None of this invalidates the claim—it bounds it—but a sound certificate is a smaller guarantee than the trajectory it protects. The reactive cost is real but uneven: the switching controller spends 1.53 vs 13.16 p.u., so

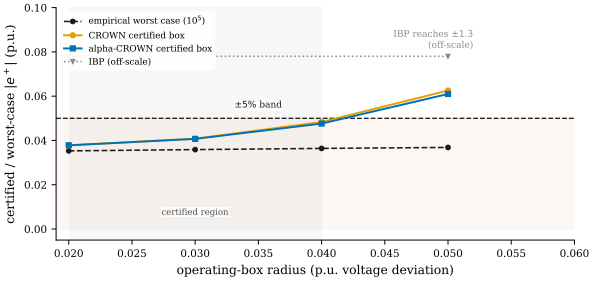


Figure 2: **The neural certificate, and where it breaks.** Certified next-step deviation from CROWN (orange) and alpha-CROWN (blue) versus the 10^5-sample empirical worst case (black, dashed), with the $\pm 5\%$ band in grey. Both methods certify the headline box r=0.04; in a fine sweep CROWN holds to 0.0409 and alpha-CROWN to 0.0415, a fraction of the true boundary 0.149. At the headline radius the certified bound is $3.22\times$(CROWN) and $3.05\times$ (alpha-CROWN) the empirical worst case. IBP reaches ± 1.3 and certifies nothing.

Table 2: Exact certificate of the switching-reference controller (single DC). The gain condition holds, and the closed-form envelope contains every trajectory.

Quantity	Symbol	Value
Gain condition $0 < K < 2X^{-1}$	–	satisfied
Droop gain	k	0.351
Spectral radius	$\rho(I - XK)$	0.99941
Contraction constant	C	48.82
Envelope utilization	$\max_j e_j / S\bar{d}$	1.26%
Scalar bound (vacuous)	$\sqrt{n}C\bar{d}/(1-\rho)$	252 p.u.

Table 3: Neural certificate and verification boundary (bus 22). *Op. box*: largest operating half-width in the headline sweep, both methods proving $r = 0.04$. *Boundary*: half-width at which each method's certificate first fails in a fine sweep. *Tight.*: certified bound over the 10^5-sample empirical worst case at $r = 0.04$. Certified one-step contraction $\gamma = 0.614$ (compute) /0.592 (comm) at box half-width 0.02.

Method	op. box	boundary	tight.@r
IBP	none (± 1.3)	–	–
CROWN	0.040	0.0409	$3.22\times$
alpha-CROWN	0.040	0.0415	$3.05\times$
empirical (worst case)	–	0.1494	–

the guarantee does not cost energy here, it saves it, because failure was the expensive thing. In the two-DC case the adaptive controller spends 8.21 p.u. and still grazes the band 0.006% of the time, which we report. IBP failing outright means the practical claim depends on the backward pass being maintained; as defense in depth, the gain condition (5) is re-checked against the raw eigenvalues and the envelope $S\bar{d}$ re-derived by direct matrix-sum accumulation (App. E).

The workload is a modeled periodic square wave (1), not a measured hardware trace; the switching structure, not the exact waveform, is what the certificate depends on. LinDistFlow (2) drops losses and holds a fixed voltage magnitude; a full AC flow would need a nonlinear certificate. The neural certificate is one-step box invariance and contraction, not multi-step reachability; chaining the certified map is the natural next step. The exact certificate is silent on the deadband and saturation the real controllers carry, and every number here is on one radial feeder. Nothing here is deployment authorization for a physical campus: protection coordination, model validation on a measured trace, and operator review all remain necessary.

5 Broader impact

in seconds, a load class growing at roughly 15% per year, more than four times faster than total electricity [9]; operators hesitate to connect such campuses because a fast-moving load can violate the voltage band even when its average draw is acceptable. Admitting either more AI load or more co-located renewables needs a certificate, not a sampled satisfaction fraction, and this paper provides it: on the linearized model (2), the exact envelope $S\bar{d}$ certifies every state the switching loop can reach, and the neural certificate fixes a box of radius 0.04 p.u. on which a learned policy provably holds all 32 buses inside $\pm 5\%$. Because a data-center step and a renewable ramp are both fast-moving injections along the same root-to-bus paths, this one certificate bounds both kinds of headroom: the radius that admits more AI load also quantifies how much intermittent generation can be co-located before the band is at risk. Two honest scopes bound the claim: the certificate is sound with respect to the linearized model (2), so only as meaningful as that model on a physical three-phase feeder, and the verified radius is one-step invariance over a defined box, not a global guarantee; on both counts the result qualifies the controller on a provable operating envelope, not a deployment.

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A Feeder model, scaling, and workload

On a radial feeder LinDistFlow [6] drops the quadratic loss terms of DistFlow, giving (2) with $R_{ij} = 2\sum_{l\in \mathrm{path}(i)\cap \mathrm{path}(j)} r_l$ and likewise X_{ij}; both are symmetric positive definite as Gram matrices of the root-to-bus paths. We build R, X from the Baran-Wu 33-bus branch data at base 10 MVA / 12.66 kV with the voltage magnitude in per unit (factor 1). This reproduces the textbook profile, whose unscaled minimum reaches 0.913 p.u. at bus 18; because the unscaled feeder is stressed on a few far laterals we scale the nominal loads by 0.5, so the base profile sits in band and the switching is the clear cause of excursions (Figure 3).

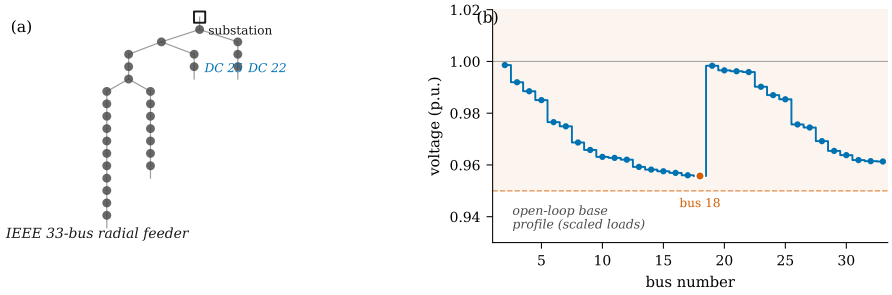


Figure 3: **Setting.** (a) IEEE 33-bus radial feeder, data-center buses 22 and 25 marked. (b) Open-loop base voltage profile under scaled nominal loads; the far-lateral minimum is 0.956 p.u. (bus 18), inside the $\pm 5\%$ band that the data-center switching then presses against.

The workload uses compute 0.40 p.u., comm 0.03 p.u., phases $60/20$ steps, and $\bar{w} = 0.006$, chosen so the fixed baseline breaches the band while the switching reference holds it [1].

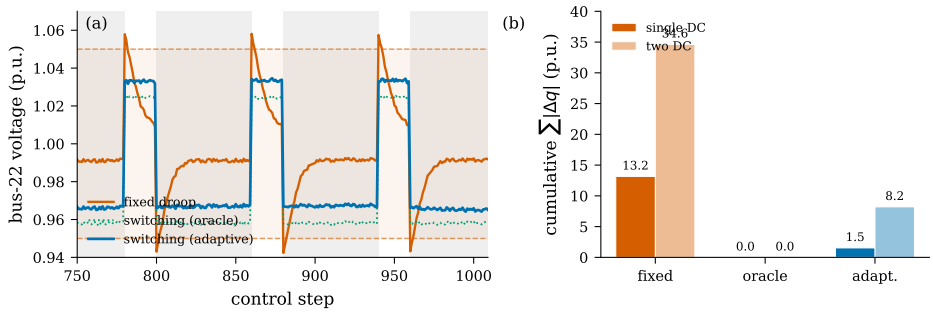


Figure 4: **The failure and the fix (bus 22).** (a) The fixed droop overshoots past $+5\%$ at every compute-to-communication transition (grey bands are compute phases) because it winds up reactive power during compute and over-corrects when the load drops; the switching reference, oracle and adaptive alike, holds the band. (b) Cumulative reactive action: the adaptive switching controller holds the band at near an order of magnitude less effort than the fixed droop (1.53 vs 13.16; 8.21 vs 34.64 at two DC, lighter bars). The oracle's zero is the ideal cancelling controller.

B Envelope proof

With (5), $\rho(A) < 1$; iterating (4) gives $e_t = A^t e_0 + \sum_{n=0}^{t-1} A^n d_{t-1-n}$, so taking elementwise norms, $|e_t| \le \sum_{n\ge 0} |A^n| \bar{d} = S\bar{d}$, convergent because $\rho(|A|) = \rho(A) < 1$ for our $A = I - XK$ [1]. Prop. 2 leaves $d_t = Rw_t$, $|w_t| \le 2\bar{w}$; we terminate the Neumann series when $\max_{ij} |A^n|_{ij} < 10^{-10}$ ($\bar{d}_{\max} = 0.00216$). The resulting envelope is shown against the empirical trajectory maximum at every bus in Figure 1 in the main text.

C Where the band breaks: a three-dimensional view

Voltage here is genuinely three-dimensional—bus, time, and magnitude—so a surface is the honest way to see where the controller fails. Figure 5 shows the per-unit deviation $v - 1$ over the whole feeder within one transition window, for the fixed droop (top) and the adaptive switching reference (bottom), with the $\pm 5\%$ band drawn as translucent planes. The fixed droop breaches the upper plane in a synchronous ridge across the downstream buses at every compute-to-communication transition, while the switching reference rides between the planes. The figure is a surface and not a projection because the third dimension—voltage height—carries the band constraint that cannot be collapsed into a line.

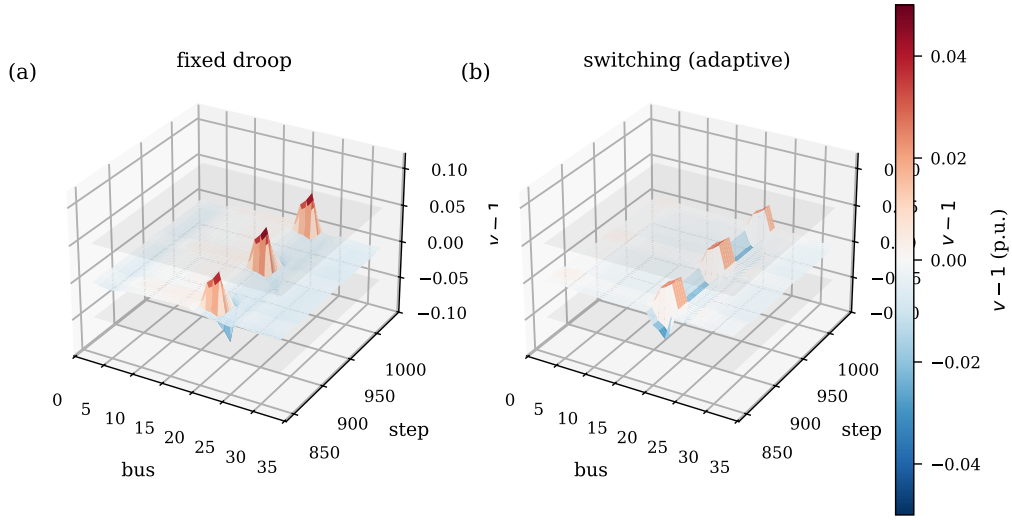


Figure 5: **The voltage surface over the feeder.** Deviation $v - 1$ across buses (axis) and one transition window (time axis); the $\pm 5\%$ band is drawn as translucent planes. Top: the fixed droop breaches the upper plane in a synchronous ridge at each compute-to-communication transition. Bottom: the adaptive switching reference holds the surface between the planes. Color encodes the signed deviation; in grayscale the breach still reads through its height.

D Controller and verification details

All controllers share (3) with a 0.01 p.u. deadband, per-step saturation $|\Delta q| \leq 0.01$, capacity $|q| \leq 0.30$, and gain $k = 0.7\,(2/\lambda_{\max}(X)) = 0.351$. The adaptive controller estimates amplitude, bias, and mode sign from a local voltage window with hysteresis and physical clipping. The neural policy is a two-hidden-layer MLP (width 32, ReLU) trained by behavior cloning plus a hinge penalty on the worst-case one-step band excursion, baked into (6). Certification uses auto-LiRPA [3]; empirical worst cases are 10^5-sample sweeps, and the bounds are sound over the whole box regardless of that sample.

E Independent checks

Each reported certificate and failure is checked by methods other than the verifier that produced it: random searches over each certified box find no feasible violation; the envelope S is re-derived by direct matrix-sum accumulation; the gain condition is re-checked against the raw eigenvalues. These give independent evidence that the successes and the failures are genuine.

F Reproducibility

Every number is produced by `make full; make figures; make summary` in the replication repository (CPU-only), which writes per-seed JSON and trajectory NPZ under `results/`; figures are regenerated from those saved files without re-simulation or retraining, and `make test` runs the feeder, controller, and gain-condition unit tests.